

Month 1 - Foundations of Programming & DSA

Topics:

- Basic Data Types & Control Structures (Variables, If-Else, Loops)
- Arrays & Strings
- Introduction to Algorithms (Time & Space Complexity)
- Basic Recursion
- Bubble Sort & Selection Sort

Resources:

- [GeeksforGeeks - Learn to Code](<https://www.geeksforgeeks.org/learn-to-code/>)
- [freeCodeCamp - Introduction to Algorithms](<https://www.freecodecamp.org/learn/javascript-algorithms-and-data-structures/>)

Practice:

- [HackerRank - Algorithm Practice](<https://www.hackerrank.com/domains/algorithms>)
- [LeetCode - Easy Problems](<https://leetcode.com/problemset/all/?difficulty=EASY>)

Month 2 - Core Data Structures & Searching/Sorting

Topics:

- Linked Lists (Singly, Doubly)
- Stacks & Queues
- Binary Search
- Insertion Sort & Merge Sort
- Introduction to Trees (Binary Trees)

Resources:

- [Programiz - Data Structures Tutorials](<https://www.programiz.com/dsa/>)

- [Coursera - Algorithms Specialization (Stanford University) - Course 1](<https://www.coursera.org/learn/algorithms-part1>)

Practice:

- [LeetCode - Easy & Medium Problems](<https://leetcode.com/problemset/all/?difficulty=EASY&difficulty;=MEDIUM>)

- [Codeforces - Div. 3/4 Problems](<https://codeforces.com/problemset/>)

Month 3 - Advanced Structures & Problem Solving

Topics:

- Trees (Binary Search Trees, AVL Trees - concepts)

- Graphs (Representations, BFS, DFS)

- Hash Tables (HashMaps)

- Quick Sort

- Basic Dynamic Programming Concepts

Resources:

- [GeeksforGeeks - Algorithms Section](<https://www.geeksforgeeks.org/algorithms/>)

- [Educative.io - Grokking the Coding Interview](<https://www.educative.io/courses/grokking-the-coding-interview-patterns-for-coding-questions>)

Practice:

- [LeetCode - Medium Problems (Focus on Patterns)](<https://leetcode.com/problemset/all/?difficulty=MEDIUM>)

- [HackerRank - Interview Preparation Kit](<https://www.hackerrank.com/interview/interview-preparation-kit>)